Rennan Seimetz Chagas



Operations Research · Optimization · Data Science

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Travel: Valid U.S. B1/B2 visitor visa

Operations research engineer and data scientist with a Ph.D. in Industrial Engineering and over a decade of experience turning complex business problems into optimization models and production software. Experience across energy, logistics, retail, and marketing, spanning mathematical modeling, data engineering, validation, deployment, and technical leadership.

PROFESSIONAL EXPERIENCE

Kpler

Sep 2025 – Present

Senior Engineer - Operations Research

Develop optimization models and production software for Kpler's Refineries product, turning refinery operations, crude logistics, and market data into forecasts of refinery runs and product output.

- Delivered complex LP/MILP models, owning mathematical formulation, data integration, testing, and production deployment across three interconnected codebases.
- Contributed to pipeline improvements that cut processing time from over five days to under six hours—more than 20× faster—through Apache Iceberg adoption, SQL optimization, and architecture reviews.
- Built Airflow workflows, schema migrations, dependency tracking, and automated monitoring, expanding ownership into data engineering and database design with Python, SQL, PostgreSQL, and AWS.
- Analyzed 2.5 million historical optimization cases through reproducible evaluation workflows, translating findings into recommendations on model quality and computational trade-offs.
- Modernized solver infrastructure with OR-Tools MathOpt/SCIP and strengthened engineering quality through code reviews, regression testing, and safer database deployments.
- Validated 1,900+ production configurations during a major rollout, confirming correct model-parameter application with zero unexplained discrepancies.
- Partnered with engineers, analysts, and product managers to turn ambiguous requirements into deliverable solutions; communicated models and technical trade-offs through presentations, visualizations, and documentation.

Petz (now Petz–Cobasi)

Dec 2024 – Sep 2025

Data Science Manager · Contractor

Led hands-on data science initiatives across retail, supply chain, operations, e-commerce, and customer analytics. Structured projects, mentored data scientists, and supported business decisions during the Petz–Cobasi merger process.

- Redesigned SKU-level demand forecasting and safety stock using seasonality, service targets, and supplier lead-time uncertainty; identified multi-million-real opportunities to reduce inventory while preserving availability.
- Developed inventory consolidation and risk-pooling analyses for merger planning, alongside optimization models for capacity-constrained staffing and shift schedules.
- Built semantic search, retrieval and reranking, recommendations, and customer analyses spanning segmentation, RFM, LTV, cohorts, and experimentation.
- Helped evolve analytical workloads across Databricks, Snowflake, and Snowpark, building reusable datasets and APIs with attention to performance and cloud cost.

PROFESSIONAL EXPERIENCE, CONTINUED

TechnipFMC · Senior Data Scientist Mar 2022 – Dec 2024
Member of the global data science team, designing models and data processes for subsea and surface businesses. Work included S&OP automation, inventory and supply chain optimization, offshore windfarm systems, AI governance, and automated document summarization with LLMs.

MINT · Senior Data Scientist Mar 2021 – Mar 2022
Designed algorithms and software within an international development team for marketing budget optimization, ROI measurement, and decision automation.

WMcCann · Lead Data Scientist Mar 2020 – Mar 2021
Applied AI/ML, optimization, and causal impact analysis to media and marketing mix modeling and campaign ROI measurement; automated analytical processes and reporting.

LORDE · Researcher / Consultant 2013 – 2022
Delivered oil and gas logistics projects, including work with Petrobras: berth scheduling, vessel routing, operations simulation, and evaluation of offshore supply policies and cargo operations.

CIRRELT · Visiting Research Scholar 2017 – 2018
Visiting researcher in operations research and logistics at CIRRELT, Montreal, Canada.

BBM Bank · Intern 2011 – 2012
Portfolio Value at Risk (VaR) measurement and process automation.

EDUCATION

Ph.D. in Industrial Engineering · Operations Research 2015 – 2022
Federal University of Rio de Janeiro (UFRJ)
Thesis: Planning offshore supply deliveries with multiproduct vessels.

M.Sc. in Industrial Engineering · Operations Research 2013 – 2015
Federal University of Rio de Janeiro (UFRJ)

B.Sc. in Industrial Engineering 2008 – 2012
Fluminense Federal University (UFF)

TECHNICAL SKILLS

Optimization	LP/MILP, mathematical programming, algorithm design, Gurobi, OR-Tools MathOpt, SCIP, PuLP
Engineering	Python, SQL, C++, Git, PostgreSQL, Airflow, Apache Iceberg, regression testing
Data & cloud	AWS, GCP, BigQuery, Snowflake, Snowpark, Databricks, dbt
Data science & AI	scikit-learn, SciPy, demand forecasting, causal analysis, LLMs, LangChain

SELECTED PUBLICATIONS

Relax-and-fix heuristics for the Multi-Commodity PSV Planning Problem with Order Selection and Time Window Decisions.
Computers & Industrial Engineering, 2023 · doi:10.1016/j.cie.2023.109040

A solution framework for the integrated periodic supply vessel planning and port scheduling in oil and gas supply logistics.
Optimization and Engineering, 2022 · doi:10.1007/s11081-022-09723-6

A solution framework for the integrated problem of cargo assignment, fleet sizing, and delivery planning in offshore logistics.
Computers & Industrial Engineering, 2021 · doi:10.1016/j.cie.2021.107653

AWARDS & LANGUAGES

ANP Award for Technological Innovation — Finalist (2018). Silver medals in the Brazilian Physics Olympiad and Rio de Janeiro Chemistry Olympiad; bronze medal in the Rio de Janeiro Mathematics Olympiad (2007).

Portuguese — Native · **English** — Fluent · **French** — Advanced · **German** — Basic